



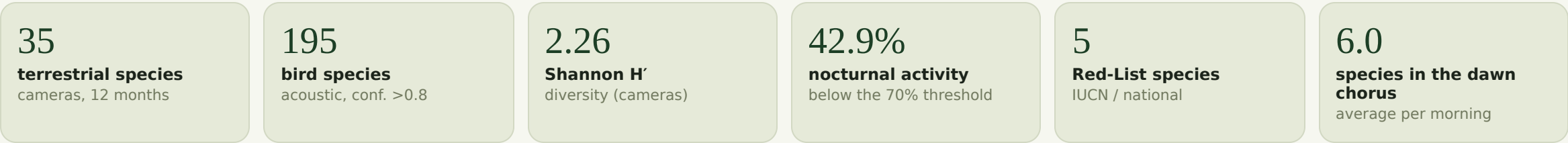
Vineyard works: *practices, rationale, calendar*

Turning 12 months of biodiversity monitoring into concrete field works — what to do, why, what data it rests on, and in which season.

Château Purcari · Ștefan Vodă PGI, Moldova Period: 29 May 2025 – 28 May 2026 (365 days) Source: Every1Counts, final report

A biodiversity-rich vineyard worth protecting

Across both monitoring methods Purcari ranks among the very top sites measured. Cameras recorded 35 terrestrial species and acoustics 195 bird species, with balanced communities. The estate already hosts a functional predator-prey web and Red-List species. The works below aim to protect the rich core, lift the poor edges and keep the vineyard a refuge for farmland wildlife without losing production.



The data behind the recommendations

■ METHODOLOGY

- 12 hotspots, each with a camera + acoustic recorder.
- Camera — motion-triggered; recorder — on a schedule.
- 365 days of collection; active hotspots ranged 6 to 10.
- Indices via bootstrap (1000 detections × 400) at equal effort.

■ KEY FINDINGS

- Shannon $H' = 2.26$, Pielou evenness $J = 0.70$ — no single species dominates.
- Turtle Dove (globally Vulnerable) — most frequent acoustically: 1428 detections.
- Nocturnal activity 42.9% — below the 70% disturbance threshold.
- Diversity is spatially structured: a stable core vs poorer edges.

The advice builds on systems already running

The estate already runs precision farming and biomonitoring — the works below extend what exists rather than starting from scratch.



Weather-based auto-irrigation

–15-30% WATER

Subsurface drip with a station that, from the weather, advises what, where and when to water.



UV tractors

–40% TREATMENTS

Irradiate the vines with UV light instead of chemical spraying.



Grass cover

Inter-row grass seeding and vine-residue return to the soil.



25 ha organic

Area under organic conversion.



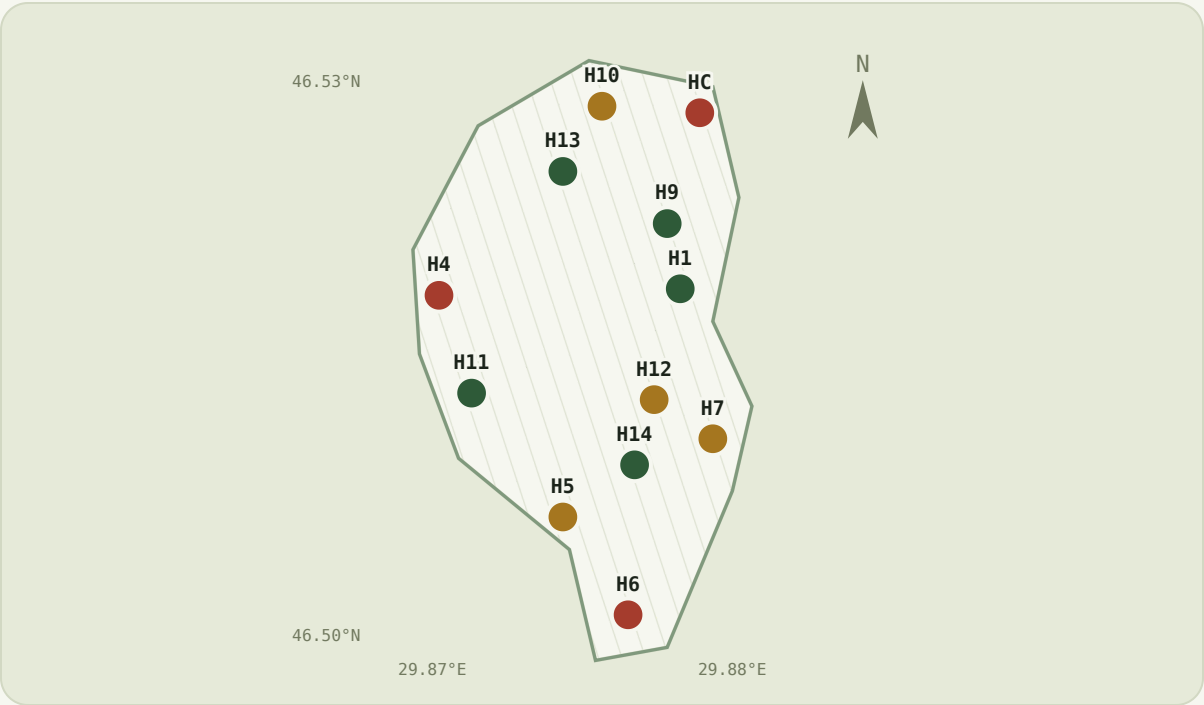
Biomonitoring network

12 hotspots: camera + acoustics, year-round.

Three zone types set where to do what

Spatial analysis and species-composition clustering split the site into a stable core, transition zones and a poor periphery. Works target zones, not the site 'in general'.

HOTSPOTS BY ZONE



North up. H1-H14, HC — monitoring stations; dot colour = zone type. Positions approximate (≈ 3 km north-south).

● CORE

Reference zones

H9 H1 H13 H11 H14

Rich, balanced — a source for recolonisation. Protect.

● TRANSITION

Functional but more uniform

H5 H7 H12 H10

Generalists dominate. Room to grow by adding structure.

● PERIPHERY

Ecologically simplified

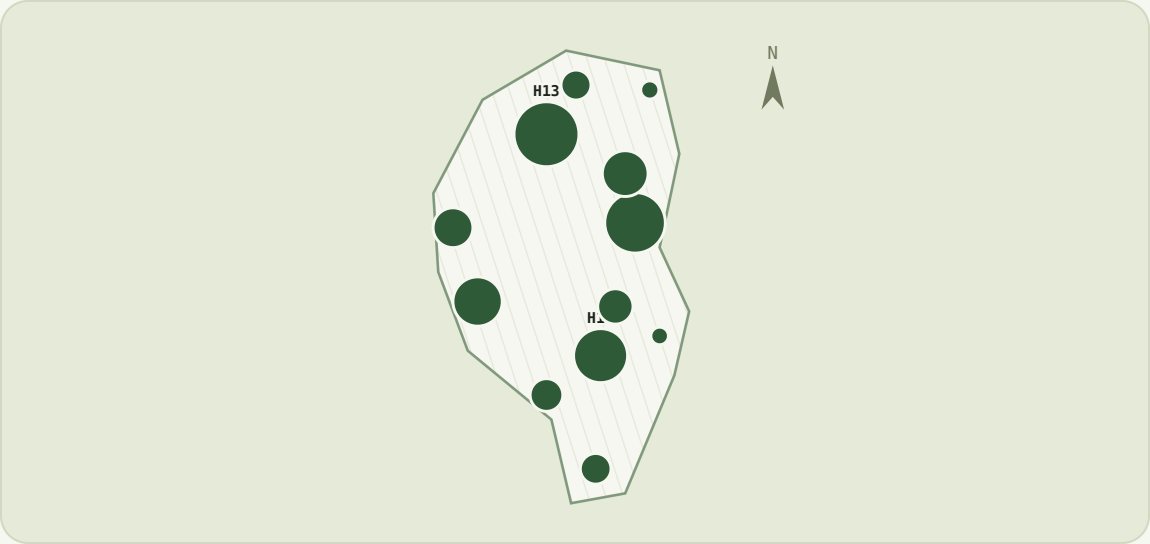
H4 H6 HC

Poorest and most isolated. The main lever for gains.

Where activity concentrates — cameras and sound diverge

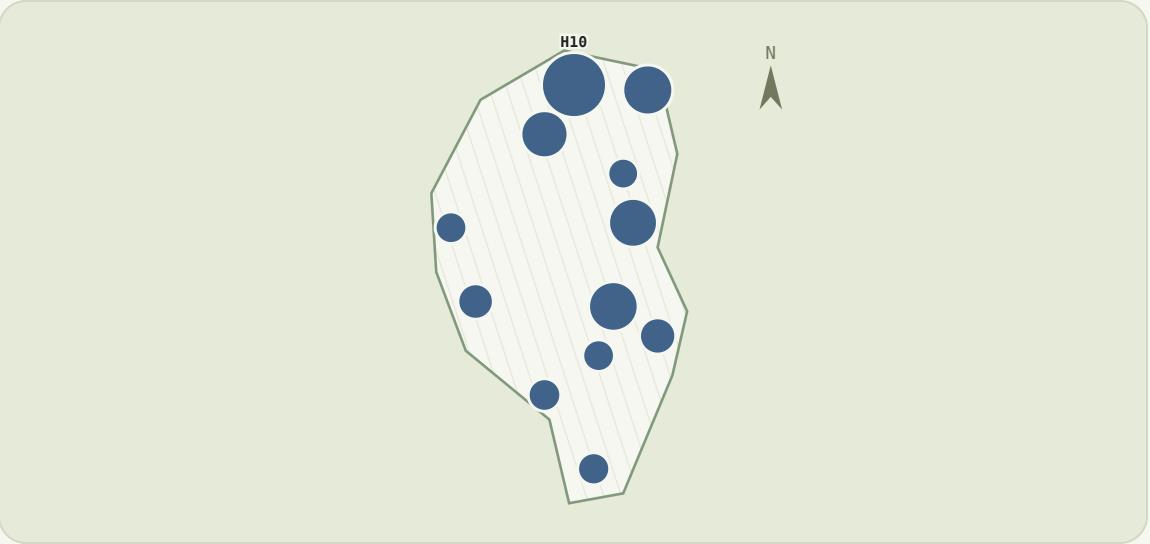
Dot size = detections at each station (from the report, approximate). Terrestrial and bird activity highlight different places: the map confirms that 'richness' depends on the taxon.

CAMERAS · TERRESTRIAL ACTIVITY



Busiest: H13, H1, H14

SOUND · BIRD ACTIVITY



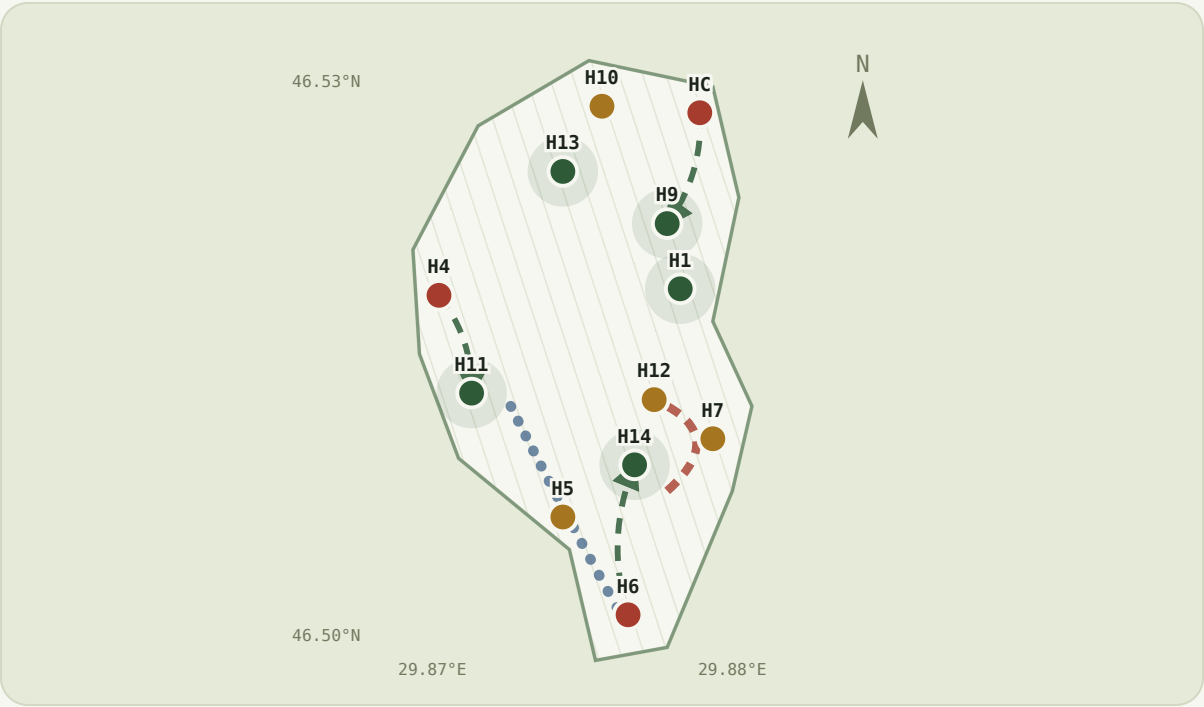
Busiest: H10, HC, H12

💡 H13, H1, H14 lead on cameras, but on sound H10 and HC come first — poor for terrestrial, rich for birds. So decisions use both methods, not one.

Where on the site each work happens

The map turns practices into space: what to protect, where corridors run, where the water buffer sits and where the boar-risk edge is.

SPATIAL WORK PLAN



North up. Green arrows — corridors from poor edges to the core; blue line — riparian buffer; red — boar-risk woodland edge.

- Protect the core** — halo around H9, H1, H13, H11, H14.
- Corridors to the core** — hedgerows from H4, H6, HC to the core (arrows).
- Riparian buffer** — no-spray strips along water (otter).
- Boar-risk edge** — monitoring and fencing at H14, H7.
- UV & irrigation — site-wide** — UV tractors and weather-based irrigation.



Protect the biodiversity core

HIGH PRIORITY

H9

H1

H13

H11

H14

Hotspots H9, H1, H13 (both methods), reinforced by H11 and H14 on cameras, form a stable rich core. These are reference zones — shield them from intensification.

■ WHY · RATIONALE

- Spatial analysis names **H9, H1, H13** as the stable core across cameras and acoustics.
- Jaccard cluster: H1/H10/H11/H12/H13/H14 share most species (H11-H14 = **0.69**).
- Kruskal-Wallis: H11, H14, H13, H1, H9 — the top group (letter A).

■ WHAT EXACTLY · AND WHEN

WINTER

Mark the surrounding woodland edges as herbicide-free buffers.

SPRING

Keep permanent grass cover and delay mowing on the adjacent inter-rows.

YEAR-ROUND

No new tracks, drainage or grubbing inside the buffer.

✓ **Benefit.** Preserves the reference zones and the source populations that recolonise the rest of the vineyard.



Restore and enrich the poor edges

HIGH PRIORITY H4 H6 H-Château

H4, H6 and H-Château are the poorest and most isolated in composition, likely fragmented. Adding structure here is the biggest lever for gains.

■ WHY · RATIONALE

- H4 and H6 — a stable poor periphery on both methods.
- H-Château, H4 and H7 — outliers, sharing very few species with the rest.
- Report: these edges are 'ecologically simplified... likely fragmented or infrastructure-influenced'.

■ WHAT EXACTLY · AND WHEN

- AUTUMN** Plant native hedgerows and shrub corridors toward the core zones.
- SPRING** Sow multi-species flower strips for pollinators and seed-eating birds.
- WINTER** Install nest boxes, insect hotels and beetle banks.

✓ **Benefit.** Raises diversity where it is lowest and stitches isolated fragments into a connected mosaic.



Safeguard the Vulnerable Turtle Dove

HIGH PRIORITY

Whole site

Hedgerows

Fallow strips

The Turtle Dove — globally Vulnerable (IUCN) — is the most frequent bird on site. Purcari is a real refuge for a declining farmland species.

■ WHY · RATIONALE

- **1428** acoustic detections of *Streptopelia turtur*, world status **VU — Vulnerable**.
- A migratory farmland species in steep decline across Europe.
- The report cites its presence as the site's 'conservation relevance'.

■ WHAT EXACTLY · AND WHEN

- SUMMER** Keep seed-bearing strips and fallow patches for summer food.
- SUMMER** Do not cut hedgerows or scrub in the nesting season (May–August).
- SPRING** Keep shallow drinking pools / puddles.
- YEAR-ROUND** Minimise herbicides near nesting hedgerows for the seed food supply.

✓ **Benefit.** Supports breeding of a globally Vulnerable species and strengthens Purcari's conservation and ESG standing.



Keep phytosanitary pressure low

HIGH PRIORITY Whole site 25 ha organic

The rich dawn chorus and low nocturnal disturbance show a lightly stressed system. UV tractors already irradiate the vines instead of spraying — worth extending.

■ WHY · RATIONALE

- UV tractors deliver **–40%** chemical treatments, irradiating with UV instead of spraying.
- Dawn chorus **6.0** species/morning — 'rich chorus, the site is alive and little disturbed'.
- Only **42.9%** nocturnal activity — below the 70% threshold.

■ WHAT EXACTLY · AND WHEN

- SPRING** Extend UV-tractor coverage to parcels beyond the current programme.
- SUMMER** Pheromone mating disruption for grape moths instead of broad-spectrum insecticides.
- SUMMER** Cut copper/sulphur where disease pressure allows; prioritise biocontrol.

✓ **Benefit.** Sustains bird and invertebrate abundance and moves toward the 25 ha organic-conversion goal.



Protect watercourses for the otter

MEDIUM PRIORITY Riparian corridors Ponds Drainage

The otter (*Lutra lutra*, Near Threatened) was recorded on site — an indicator of healthy water. Weather-based auto-irrigation already advises what/where/when to water; its schedule helps keep spray drift away from water.

■ WHY · RATIONALE

- 3 detections of *Lutra lutra*, world status **NT — Near Threatened**.
- The otter's presence signals good water quality.
- Weather irrigation and subsurface drip cut water by **–15-30%** and reduce runoff.

■ WHAT EXACTLY · AND WHEN

- YEAR-ROUND** Keep uncultivated buffer strips along all watercourses and ponds.
- SUMMER** Align the irrigation schedule with the weather to avoid drift and runoff toward water.
- SPRING** Extend subsurface drip to reduce runoff and erosion.

✓ **Benefit.** Preserves aquatic habitat quality and the otter as a clean-water bioindicator.



Manage wild-boar damage

MEDIUM PRIORITY

H14

H7

Wood-edge parcels

Wild boar are regularly present (102 detections). Part of the food web, but they can damage vines, grapes and soil — especially where the vineyard meets the forest (H14, H7).

■ WHY · RATIONALE

- **102** wild-boar detections on cameras over a year.
- H14 and H7 sit on the south-east forest edge, where boar pressure concentrates.
- Both predators and herbivores present — a functional but stressed system.

■ WHAT EXACTLY · AND WHEN

- AUTUMN** Monitor damage on wood-edge parcels toward harvest and log incidents.
- SUMMER** Targeted electric fencing on vulnerable parcels before véraison.
- AUTUMN** Remove attractants (fallen fruit, pomace) from field edges.

✓ **Benefit.** Limits crop and soil damage while keeping boar part of the ecosystem rather than removing it.



A wildlife-friendly mowing regime

MEDIUM PRIORITY

Inter-rows

Grassed alleys

Edges

Camera data are dominated by ground species — pheasant, hare — alongside ground-nesting birds and invertebrates. How and when the inter-rows are mown affects them directly.

■ WHY · RATIONALE

- Pheasant (**1171**) and hare (**818**) — the two most frequent species on cameras.
- The estate already practises grass seeding and vine-residue return.
- Activity is continuous through the day — typical of the agro-forestry mosaic.

■ WHAT EXACTLY · AND WHEN

SPRING

Mow alternate inter-rows, leaving standing refuge strips.

SPRING

Delay first mowing until after the ground-nesting peak (late spring).

SUMMER

Mow from the centre outward so animals escape to the edges.

✓ **Benefit.** Protects hares, ground-nesting birds and pollinators while keeping the cover-crop agronomy.



Protect and continue the monitoring network

MEDIUM PRIORITY

All hotspots

Core first

Equipment losses and sensor failures interrupted collection. A secured, continuous network turns this baseline into a multi-year trend and supports organic / regenerative certification.

WHY · RATIONALE

- Repeated 'equipment disappearance' and the H9 acoustic failure (from 17.03.26).
- Monitoring was 'discontinuous', active hotspots ranging 6 to 10.
- 12 hotspots surveyed — but only after relocating the original 10.

WHAT EXACTLY · AND WHEN

- YEAR-ROUND** Anti-theft mounts and discreet placement, core first.
- SPRING** Redundancy (a spare camera/recorder) at reference zones H9, H1, H13.
- YEAR-ROUND** Regular sensor checks to catch failures like H9 early.

✓ **Benefit.** Gives a reliable yearly biodiversity trend and an evidence base for certification and ESG reporting.



Ease human pressure on the diurnal edges

LOW PRIORITY

H7

H-Château

H4

H7, H-Château and H4 skew strongly to daytime (>80% diurnal detections). This may reflect open habitat, but also human/vehicle activity that pushes fauna away.

■ WHY · RATIONALE

- Diurnal share of detections: H7 **83.3%**, H-Château **85.7%**, H4 **89.7%**.
- Report: human pressure can heighten daytime activity on edge parcels.
- The same parcels — outliers with the fewest shared species.

■ WHAT EXACTLY · AND WHEN

- SUMMER** Concentrate machinery passes into time windows, leaving quiet periods.
- AUTUMN** Add screening vegetation between tracks/buildings and parcels.
- YEAR-ROUND** Reduce or shield night lighting at the château and access roads.

✓ **Benefit.** Restores habitat value and nocturnal access for wary species on the busiest edges.

All works by season

28 concrete works spread across the seasons — a working schedule for planning the year.

 Spring 8 • Permanent grass cover at the core, delay mowing • Multi-species flower strips in the inter-rows • Shallow drinking pools / puddles for the dove • Extend subsurface drip (weather irrigation) • Extend UV-tractor coverage to new parcels • Mow alternate rows, leaving refuge strips • Delay first mowing until after nesting • Equipment redundancy at H9, H1, H13	 Summer 8 • Seed-bearing strips and fallow for food • Do not cut hedgerows/scrub (May–August) • Align irrigation with weather, keep spray off water • Targeted electric fencing before véraison • Pheromone mating disruption vs insecticides • Cut copper/sulphur, prioritise biocontrol • Mow from the centre outward • Machinery passes in time windows	 Autumn 4 • Hedgerows / shrub corridors toward the core • Monitor wild-boar damage on wood edges • Remove attractants — fallen fruit, pomace • Screening vegetation by tracks and buildings	 Winter 2 • Woodland edges at the core — herbicide-free buffers • Nest boxes, insect hotels, beetle banks	 Year-round 6 • No new tracks, drainage or grubbing in the buffer • Minimal herbicides near nesting hedgerows • Uncultivated buffer strips along water • Reduce/shield night lighting • Anti-theft equipment mounts • Regular sensor health checks
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Rollout order by priority

9 practices ranked by biodiversity impact and urgency. Start with high priority, then medium; the edge — when possible.

FIRST

High

Protect the core

Restore the edges

Safeguard the Turtle Dove

Low phytosanitary pressure (UV tractors)

THEN

Medium

Protect watercourses (otter)

Manage wild boar

Wildlife-friendly mowing

Protect the monitoring network

WHEN POSSIBLE

Low

Ease pressure on diurnal edges

Keep the core, lift the edges, keep watching



Protect what's already valuable. The H9/H1/H13/H11/H14 core is the reference; no intensification here.



The flagship is the Turtle Dove. Simple measures support a globally Vulnerable species and strengthen ESG.



Irrigate smartly. Weather-based auto-irrigation advises what/where/when — saving water and cutting runoff to watercourses.



Improve where it's poorest. Hedgerows and corridors on H4/H6/H-Château give the biggest gain.



Keep pressure low. UV tractors, biocontrol and pheromones preserve the rich chorus and the path to 25 ha organic.



Keep monitoring. A secured network gives a multi-year trend and a basis for certification.

Based on: Every1Counts – 'Analysis of data collected over 12 months on the Château Purcari site', final report, 08.07.2026.